

Homework — 2.2.2 Computational Methods — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] — Any two (1 each): can be **decomposed** into sub-problems; has clear **inputs, processes and outputs**; is repetitive / rule-based (algorithmic); allows useful **abstractions**; data can be **structured/modelled**; has a clear, **finite** sequence of steps.
2. [2] — 1 mark: **decomposition** breaks a problem into smaller, manageable sub-problems that may be **different** sub-tasks. 1 mark: **divide and conquer** repeatedly splits a problem into **smaller instances of the same problem**, solves the parts, then **combines** the results (e.g. binary search, merge sort). (*Key contrast: different sub-tasks vs same-type smaller instances combined.*)
3. - (a) [1] — **Visualisation**. - (b) [1] — Any one: patterns/trends/correlations/outliers are easier to spot than in raw figures; aids faster decision-making; communicates findings more clearly.
4. - (a) [1] — A **heuristic** is a rule of thumb that gives a **good-enough (not guaranteed optimal)** answer quickly. - (b) [2] — 1 mark: an exact method must check an enormous number of route combinations — the work grows factorially, so it is **intractable** / too slow for many locations. 1 mark: a heuristic finds a **good-enough** route **quickly**, accepting that it may **not be optimal**, which is acceptable in practice.
5. [2] — 1 mark: pipelining splits a process into **stages** so that different stages can work on **different items at the same time**, increasing throughput. 1 mark: valid example, e.g. CPU **fetch–decode–execute** (while one instruction is decoded, the next is fetched); or an assembly line.
6. [2] — 1 mark: **representational abstraction** removes/hides detail not relevant to the problem, keeping only what is needed — e.g. the London Tube map keeps line connections but drops true distances/geography. 1 mark: **abstraction by generalisation** groups things by **shared characteristics** so they can be treated the same — e.g. classifying many shapes as "polygon", or grouping car types as "vehicle". (*Each needs the idea + an example.*)

Part B (6 marks)

7. [1] — A **function** returns a **single value** (usable in an expression); a procedure returns no value.
8. [2] — 1 mark: **stack overflow**. 1 mark: with no reachable base case the subroutine keeps calling itself, each call **pushing a new stack frame**, until memory is exhausted.
9. [3] — 1 mark: **thinking ahead** = identifying in advance the inputs and outputs / anticipating the needs of a solution before/while designing it. 1 mark: identifying **inputs and outputs** required (or reusable components / caching / preconditions). 1 mark: valid example — e.g. determining what data must be captured by a form before building it; **caching** likely-needed web resources; pre-fetching in a CPU pipeline. (*Award the example mark only with a clear, relevant case.*)

Total: 14 + 6 = 20